

CSE 421 Assignment 3

Name: Mehedi Hasan Rowdru

ID: 24101226 ; Section: 25

Date

S S M T W T F

Answer 1

Given,

2 Networks X and Y

X

• 510 Hosts

• First usable ip: 192.168.20.1

Source: 192.168.20.1

Dest: 192.168.23.255

i

Host Name	Host req	+2	2^n IP Block size	Host Bits	Network Bits
X	510	512	512	9	23

Subnet = /23

Network X:

Network: 192.168.20.0/23

Range: 192.168.20.0/23 - 192.168.21.255/23

Network Y

start = End of X + 1

= 192.168.22.0/23

∴ Network of Y: 192.168.22.0

(Ans)

X

Start:

.00010100.00000000

= 20.0

End:

.00010101.11111111

= 21.255

Y

+1

Start: .00010110.00000000

= 22.0

ii Block size = 512 [From "i"]

∴ Max num. of host Network Y supports = $512 - 2 = 510$ (Ans)

iii Given IP Block 190.132.0.0/19

R1 LAN → 2048 Hosts

R2 LAN → 120 "

Switch LAN → 100 " (Assume)

Server LAN → 20 " (Assume)

Router Links → 2 " each (assume)

Hosts	No of Hosts	+2	IP Block	Host Bits	Network Bits	Network Add	Broadcast Add	Usable Range
R1	2048	2050	4096	12	20	190.132.0.0/20	190.132.15.255	190.132.0.1 - 190.132.15.254
R2	120	122	128	7	25	190.132.16.0/25	190.132.16.127	190.132.16.1 - 190.132.16.126
Switch	100	102	128	7	25	190.132.16.128/25	190.132.16.255	190.132.16.129 - 190.132.16.254
Server	20	22	32	5	27	190.132.17.0/27	190.132.17.31	190.132.17.0 - 190.132.17.30
RL1	2	4	4	2	30	190.132.17.32/30	190.132.17.35	190.132.17.33 - 190.132.17.34
RL2	2	4	4	2	30	190.132.17.36/30	190.132.17.39	190.132.17.35 - 190.132.17.38
RL3	2	4	4	2	30	190.132.17.40/30	190.132.17.43	190.132.17.41 - 190.132.17.42

Calculation Next Page.

R1

Start: .00000000.00000000

End: .00001111.11111111

=.15.255

R2

Start: .00010000.00000000

=.16.0

End: .00010000.01111111

=.16.127

Switch

Start: .00010000.10000000

=.16.128

End: .00010000.11111111

=.16.255

Server

Start: .00010001.00000000

=.17.0

End: .00010001.00011111

=.17.31

RL1

S: .00010001.00100000

=.17.32

e: .00010001.00100011

=.17.35

RL2

S: .00010001.00100100

=.17.36

e: .00010001.00100111

=.17.39

RL3

S: .00010001.00101000

=.17.40

e: .00010001.00101011

=.17.43

Answer 2

Given,

Broadcast Address: 7.16.255.255/18

R2 LAN $\rightarrow$ 2500 HostsSwitch LAN $\rightarrow$ 125 "R4 LAN $\rightarrow$ 64 "Router Links (R1-R2, R2-R3, R1-SW, SW-R4) $\rightarrow$ 2 hosts each

1/2.

Subnet: /18

Network Bits = 18

Host Bits = $32 - 18 = 14$

Broadcast in Binary: 00000111.00010000.11111111.11111111

Network Address in Binary: 00000111.00010000.11000000.00000000

 $\therefore$ Network Address = 7.16.192.0

(Answer)

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Host	No. of Host	+2	IP Block	Host Bits	Net Bits	Network Add	Broadcast Add	Usable Range
R2 LAN	2500	2502	4096	12	20	7.16.192.0/20	7.16.207.255	7.16.192.1 - 7.16.207.254
Switch LAN	125	127	128	7	25	7.16.208.0/25	7.16.208.127	7.16.208.1 - 7.16.208.126
R4 LAN	64	66	128	7	25	7.16.208.128/25	7.16.208.255	7.16.208.129 - 7.16.208.254
R1-R2	2	4	4	2	30	7.16.209.0/30	7.16.209.3	7.16.209.1 - 7.16.209.2
R2-R3	2	4	4	2	30	7.16.209.4/30	7.16.209.7	7.16.209.5 - 7.16.209.6
R1-SW	2	4	4	2	30	7.16.209.8/30	7.16.209.11	7.16.209.9 - 7.16.209.10
SW-R4	2	4	4	2	30	7.16.209.12/30	7.16.209.15	7.16.209.13 - 7.16.209.14

R2 LANS: ~~7.16.192.0~~ 00010000.11000000.00000000

E: 00011111.11111111.11111111

R2 LAN

S: 11000000.00000000

E: 11001111.11111111

= 207.255

Switch LAN

S: 11010000.00000000

= 208.0

E: 11010000.01111111

= 208.127

R4 LANS: 11010000.10000000
= 208.128E: 11010000.11111111
= 208.255R1-R2S: 11010001.00000000
= 209.0E: 11010001.00000011
= 209.3R2-R3

S: 11010001.00000100 = 209.4

E: 11010001.00000111 = 209.7

R1-SW

S: 11010001.00001000 = 209.8

E: 11010001.00001011 = 209.11

SW-R4

S: 11010001.00001100 = 209.12

E: 11010001.00001111 = 209.15

c

For R4 LAN,

$$\text{Subnet} = /25$$

$$\text{Host Bits} = 32 - 25 = 7$$

$$\text{Usable} = 2^7 - 2 = 126 > \text{Required} = 64$$

$$\therefore \text{Waste} = 126 - 64 = 62$$

(Ans)